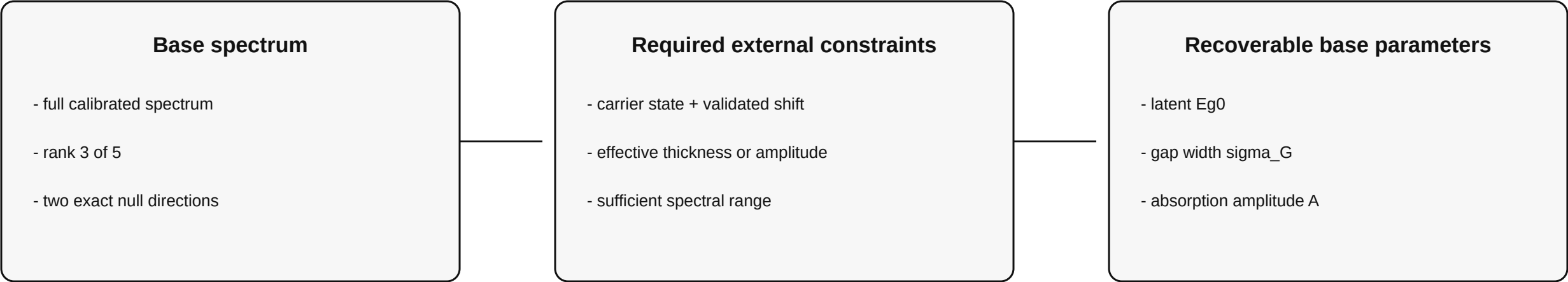


Figure 7. Measurement design and external-validation boundary

Exact identifiability requirements do not depend on recruiting a collaborator



Validation status

Component	Analytical	Numerical	Source reproduction	Real-spectrum validation
Transition distribution	yes	yes	n/a	no
Herrmann tail	yes	yes	yes	no new specimen
Chang cutoff	yes	yes	source-bounded	no
Carrier filling	yes	yes	regime anchor	no
Unified theorem	yes	yes	n/a	no

ANALYTICAL CORE COMPLETE; EXTERNAL SPECIMEN VALIDATION REMAINS OPEN